

The general-domain Taylor-centre conjecture of arXiv:1710.03114 was answered in 2019

Literature-status note for `fa_banach_001`

17 August 2026

Status. `literature_already_answered` (full conjecture, affirmative).

Original conjecture

Let $\Omega \subset \mathbb{C}$ be a simply connected domain with $0 \notin \Omega$, and for $f \in \mathcal{H}(\Omega)$ put

$$T_N f(z) = \sum_{n=0}^N \frac{f^{(n)}(z)}{n!} (-z)^n.$$

The abstract and Section 1 of Panagiotis [1] record Nestoridis's conjecture that

$$S(\Omega) = \{f \in \mathcal{H}(\Omega) : \{T_N f : N \geq 0\} \text{ is dense in } \mathcal{H}(\Omega)\}$$

is a dense G_δ subset of $\mathcal{H}(\Omega)$. That paper proves the claim when Ω is a disc avoiding the origin, leaving the stated general simply connected case open.

Exact later answer

Bernal-González, Cabana-Méndez, Muñoz-Fernández, and Seoane-Sepúlveda [2] prove the full conjecture and more. Their Theorem 2.10 (paper page 8; journal page 487) states that, for every simply connected domain $G \subset \mathbb{C}$, the following are equivalent:

1. $0 \notin G$;
2. the sequence (T_N) is universal on $\mathcal{H}(G)$;
3. (T_N) is mixing;
4. $S(G)$ is residual in $\mathcal{H}(G)$;
5. $S(G)$ is dense-lineable in $\mathcal{H}(G)$.

The same paper recalls that $S(G)$ is always a G_δ set; combined with the residuality assertion, this gives precisely the dense G_δ conclusion requested in arXiv:1710.03114. Thus the answer is affirmative for every simply connected domain avoiding zero, with the stronger mixing and dense-lineability conclusions.

The answering paper explicitly identifies Panagiotis's disc theorem in its introduction and then announces that it will characterize non-vacuity for all simply connected domains. This is therefore a direct later answer, not an indirect consequence inferred from unrelated machinery.

Independent route found during triage

A direct attack before the final novelty check found a short alternative coordinate picture. For a branch $L = \log$ on Ω , conjugation by $f \mapsto f \circ \exp$ sends T_N to

$$A_N = \prod_{j=1}^N \left(1 - \frac{D}{j}\right), \quad D = \frac{d}{dw}.$$

On $e^{\lambda w}$, the eigenvalue is $\prod_{j=1}^N (1 - \lambda/j)$, which tends to zero for $\operatorname{Re} \lambda > 0$ and has modulus tending to infinity for $\operatorname{Re} \lambda < 0$. Exponentials from either half-plane have dense span in $\mathcal{H}(L(\Omega))$, giving the usual universality criterion. This corroborates the affirmative theorem but is not promoted as a new result because Theorem 2.10 already settles the mathematical question.

Verification and scope

The match was checked at the level of the operator formula, the domain hypotheses, and the conclusion. The original conjecture appears in the abstract and Section 1 of arXiv:1710.03114. The exact answer is Theorem 2.10 of the 2019 paper. Its Question 2.11 asks the strictly stronger and separate question whether $S(G)$ is spaceable; no claim about that question is made here.

References

- [1] C. Panagiotis, *Universal partial sums of Taylor series as functions of the centre of expansion*, Complex Variables and Elliptic Equations (2019), [doi:10.1080/17476933.2019.1611791](https://doi.org/10.1080/17476933.2019.1611791); [arXiv:1710.03114](https://arxiv.org/abs/1710.03114).
- [2] L. Bernal-González, H. J. Cabana-Méndez, G. A. Muñoz-Fernández, and J. B. Seoane-Sepúlveda, *Universality of sequences of operators related to Taylor series*, J. Math. Anal. Appl. **474** (2019), 480–491, [doi:10.1016/j.jmaa.2019.01.056](https://doi.org/10.1016/j.jmaa.2019.01.056).